SUPPORTING ADULT GUIDANCE

Forming an equation

Equations, Inequalities and Simultaneous Relationships · Lesson 4. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson is about **forming** equations from situations — worded 'think of a number' puzzles, cost and age problems, and simple geometric ones — then solving and, crucially, **interpreting** the answer back in context. It draws together the solving methods of Lessons 1–3 and adds the modelling skill of translating a situation into algebra and back.

Driving Question: *How do I turn a situation into an equation I can solve?*

The mathematics, in plain terms

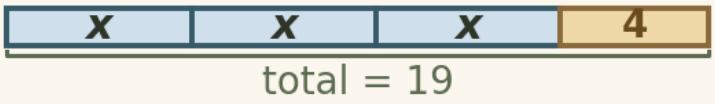
To model a situation, name the unknown quantity as x and express each stated relationship in terms of it, giving an equation. Solving the equation yields x; interpreting means expressing the quantity the question actually asks for (which may be x, or $x + 4$, or $2x$, etc.) and confirming it satisfies the original wording. A bar (tape) model is a useful bridge: it lays the parts and whole out visually so the equation can be read straight off.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Think of a number

'I think of a number, multiply it by 3, then add 4, and the result is 19.' The bar shows 3 x-blocks and a 4-block making 19.

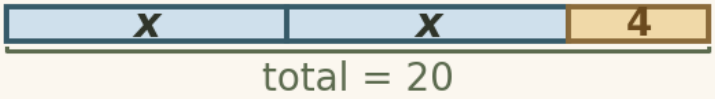


1. Let the number be x, and write the situation.	$3x + 4 = 19$
2. Subtract 4 from both sides.	$3x = 15$
3. Divide both sides by 3.	$x = 5$
4. Check against the story.	$5 \times 3 = 15$, then $+ 4 = 19$ ✓

$x = 5$ — form $3x + 4 = 19$, then solve; $5 \times 3 + 4 = 19$ matches the story.

Example 2 — Interpreting the answer

'Emma is x; her sister is 4 years older; together they are 20.' The bar shows two x-blocks and a 4-block making 20.



1. Write the total.	$x + (x + 4) = 20$, so $2x + 4 = 20$
2. Subtract 4 from both sides.	$2x = 16$
3. Divide both sides by 2.	$x = 8$
4. Interpret: the question may want the sister.	Emma 8, sister 8 + 4 = 12 ✓

Emma is 8 (and her sister 12) — solving gives x = 8; interpreting turns x into each age. Check: 8 + 12 = 20.

Name, write, solve, interpret, check

Encourage the same sequence every time: name the unknown, write the equation phrase by phrase, solve it, then interpret the answer in the situation and check it fits. The two hardest steps are the first (deciding what x stands for) and the last (turning x back into the answer the question wants) — give them time, and use a bar model when a situation is hard to picture.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Translate a worded situation into an equation, term by term.	~12 min
Movement	Read an equation straight off a bar (tape) model, then solve it.	~10 min
Reflection	Solve, then interpret — answer what the situation actually asked.	~10 min
Creativity	Build your own 'think of a number' puzzle from a chosen answer.	~8 min
Rest	A short pause on the power of naming the unknown.	~2 min
Nutrition	Contextual (Mode A): forming equations for real budgeting and planning.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Not defining the unknown, so the algebra floats free.
Insist on 'let x be…' first. Ask: 'What exactly does x stand for here — in words?'

Mistranslating order, e.g. writing 3 + 4x for 'multiply by 3, add 4'.
Walk the sentence left to right: 'What happens to the number first? Then what?' Match each phrase to a term.

Giving x when the question asked for something else (the sister's age, the length).
Re-read the question after solving: 'You found x. Does the question want x, or something built from it?'

Not checking against the situation.
Make it routine: 'Put your answer back into the words — does the story come out true?'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: writing expressions, solving two-step equations, and interpreting an answer in context. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner writes 3 + 4x for 'multiply by 3 then add 4', or answers x = 8 when the question asked for the sister's age. The aim is a question that returns the thinking.

- "What does x stand for here — can you say it in words?"
- "What happens to the number first? So what is the first term? Then what is added?"

- “You found x = 8. Read the question again — does it want x, or the sister's age?”
- “Put your answer back into the story — does it come out true?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

How do learners decide what x should be?

Usually x is the quantity the question asks about, or the simplest unknown the others can be written from (e.g. the younger age, the width). If a choice makes the algebra simpler, take it.

Are bar models necessary?

No, but they help enormously when a situation is hard to picture — they lay the parts and whole out so the equation can be read straight off. Offer one; don't require it.

Why stress interpreting so much?

Because in real problems the solution to the equation is rarely the final answer. Interpreting — and checking against the situation — is what makes the maths actually answer the question.

Do answers have to be whole numbers?

Not necessarily — the stretch question gives x = 2.5. Whether a non-whole answer makes sense depends on the context (2.5 miles is fine; 2.5 people is not), which is itself part of interpreting.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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